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MILITARY SPECIFICATION

CALCIUM RESINATE

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers the requirements of calcium resinate.

1.2 Classification. Calcium resinate shall be of the following types, as specified (see 6.2).

Type I — Fused.

Type II — Precipitated.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATION

FEDERAL

RR-S-366 — Sieves, Standard, for Testing Purposes.

STANDARD

MILITARY

MIL-STD-129 — Marking for Shipment and Storage.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Type I calcium resinate shall be yellow or brown in color. Type II calcium resinate shall be a yellowish white amorphous powder, free from lumps, cakes and agglomerations.

3.2 Composition. The chemical and physical properties shall conform to the requirements in table I.

FSC 6810

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TABLE I. *Chemical and physical properties*

	Type I	Type II
Calcium resinate content, percent	56.0 min	77.0 min
Acid number	64 max	33 max
Chloroform-insoluble matter, percent	2.0 max	3.0 max
Water content, percent	2.0 max	5.0 max
Flame test	Yellow-red	Yellow-red
Granulation through 60 sieve, percent	none.	75 min
Granulation through 4 sieve, percent	90.	

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.2 **Lot.** A lot shall consist of material from the same batch or blending operation and shall consist of material subjected to the same processing operations and conditions. In no case shall a lot exceed 10,000 pounds.

4.3 **Sampling.** Proper sampling of the calcium resinate is highly important. Select 10 percent, but in no case more than 10 of the containers comprising the lot. Remove from each selected container sufficient amounts of the material to form a primary sample of about 8 ounces. Mix this sample thoroughly, place a 4 ounce portion in a stoppered bottle (the air space at the top should be kept at an absolute minimum) and label each primary sample so that the container from which it was taken can be easily identified. Mix the remaining portions of the primary samples thoroughly and quarter until a composite sample of about 4 ounces is obtained. Place the composite sample in a stoppered bottle (the air space at the top should be kept at an absolute minimum) and label so as to show the name of the material, manufacturer, purchase order, and number of pounds in the lot. All acceptance tests shall be made on the composite sample representative of the lot. Hold the primary samples for possible future examination should the composite sample fail to meet the requirements. If at any time it becomes apparent during sampling that the batch is not uniform, the inspector may require that additional samples be taken and tested for compliance with the requirements of this specification.

4.4 **Tests.**

4.4.1 *Acid number.* Transfer approximately 2 grams of the sample weighed to 1 milligram, to a 250-milliliter Erlenmeyer flask, add 25 milliliters of chloroform, and allow to stand for 1 hour with occasional swirling. Add 25 milliliters of 95 percent ethyl alcohol and 2 milliliters of phenolphthalein solution (made by dissolving 0.1 gram phenolphthalein in 100 milliliters of a mixture of equal parts of water and alcohol). Titrate with a standard 0.1 N sodium hydroxide solution to a definite pink end point. Deduct a blank obtained by titrating a solution containing 25 milliliters of chloroform, 25 milliliters of alcohol, and 2 milliliters of phenolphthalein solution. Calculate as follows:

$$\text{Acid No.} = \frac{\text{ml of NaOH} \times \text{Norm. of NaOH} \times 56.1}{\text{gm of sample}}$$

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4.4.2 *Chloroform-insoluble matter.*

4.4.2.1 *Type I.* Transfer approximately 2 grams of the sample accurately weighed to 0.1 milligram, to a 250-milliliter beaker, add 50 milliliters of chloroform, and allow to stand for 1 hour, while swirling occasionally. Filter through a Gooch crucible prepared with Powminco acid washed Gooch asbestos (see 6.3) and transfer and wash the precipitate with chloroform. Save the filtrate for the calcium resinate determination. Dry the crucible at 105° C. for 1 hour, cool, and weigh. Calculate as follows:

$$\text{Percent chloroform-insoluble matter} = \frac{\text{gm of residue}}{\text{gm of sample}} \times 100$$

4.4.2.2 *Type II.* Proceed as with type I but use a 1 gram sample.

4.4.3 *Calcium resinate content.*

4.4.3.1 *Reagents.*

Buffer solution. Dissolve 67 grams of ammonium chloride in about 400 milliliters water, add 570 milliliters of ammonium hydroxide, and dilute to 1 liter with water.

Eriochrome Black T indicator. Dissolve 0.4 grams of Eriochrome Black T in 100 milliliters of methyl alcohol.

Standard calcium solution. Dissolve 2.20 grams of reagent grade calcium carbonate in a mixture of 20 milliliters of water and 5 milliliters of hydrochloric acid by boiling gently on the hot plate. Cool and dilute to 2 liters. Standardize as follows: Pipet a 100 milliliter aliquot into a 400 milliliter beaker, and add 100 milliliters of water and 5 milliliters of hydrocloric acid. Heat to about 95° C. and add 50 milliliters of oxalic acid solution (2 percent) that has also been heated to about 95°. Add ammonium hydroxide dropwise with stirring until the solution smells faintly of ammonia. Place on the bench and allow to stand for 2 hours. Filter through a filtered glass crucible of medium porosity. Transfer the precipitate to the crucible and wash with 0.1 percent oxalic acid solution. Finally wash the precipitate once with a small amount of water. Place the crucible into a 600 milliliter beaker containing 300 milliliters of water and 12 milliliters of sulfuric acid. Stir, heat to about 60° to dissolve and titrate with 0.1 N potassium permanganate. Calculate as follows:

$$\text{Gms. of } CaCo_3 \text{ per ml of solution} = \frac{\text{ml of } KMnO_4 \times \text{Norm. of } KMnO_4 \times 0.050045}{\text{ml of } CaCo_3 \text{ solution taken for titration}}$$

Versenate solution. Dissolve 8 grams of Versenate (disodium-Dihydrogenethylenediaminetetraacetate) and 0.2 grams of $MgCl_2\ 6H_20$ in water and dilute to 2 liters. Standardize as follows: Pipet a 20 milliliter aliquot of the standardized $CaCo_3$ solution into a 250 milliliter Erlenmeyer flask, and add 30 milliliters of water, 10 milliliters of buffer solution and 5 drops of Eriochrome Black T indicator. Titrate with Versenate to a bright blue color that does not change on the further addition of Versenate. Calculate as follows:

$$\text{Factor (gm of calcium resinate per ml of Versenate solution)} = \frac{\text{ml of } CaCO_3 \text{ solution} \times \text{gm } CaCO_3 \text{ per ml} \times 6.4237}{\text{ml of Versenate solution taken for titration}}$$

4.4.3.2 *Procedure.*

4.4.3.2.1 *Type I.* Transfer the filtrate from the chloroform-Insoluble matter to a 400 milliliter beaker and evaporate to dryness by heating on the steam bath. Add 35 mill-

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iliters of nitric acid and cover with a watch glass. Boil down to dryness by heating on a hot plate at moderate heat, and heat at dryness for a few minutes until the residue is charred. This charring is important. Allow to cool a few minutes, add 35 milliliters more of nitric acid and again evaporate till charring. Allow to cool a few minutes, and add 35 milliliters of nitric acid and 8 milliliters of perchloric acid. Cover with a watch glass, evaporate to fumes of perchloric acid and continue the fuming until the volume is about 1 milliliter. Open the cover lid and evaporate to dryness. Allow to cool a few minutes, dissolve the salts in water, and dilute to 250 milliliters in a volumetric flask. Pipet a 50 milliliter aliquot (equivalent to 0.4 gram sample) into a 250 milliliter Erlenmeyer flask, and add 10 milliliters of buffer solution and 5 drops of Eriochrome Black T indicator. Titrate with the Versenate solution to a bright blue color that does not change on the further addition of the Versenate.

Calculate as follows:

$$\text{Percent calcium resinate} = \frac{\text{ml of Versenate solution x factor x 100}}{\text{gm of sample in aliquot}}$$

4.4.3.2.2 *Type II.* Proceed with type I but use a 100 milliliter aliquot (equivalent to 0.4 gram sample).

4.4.4 *Water content.*

4.4.4.1 *Type I.* Transfer a 100 gram sample to a dry 500=milliliter balloon flask and add milliliters of reagent grade chloroform. Attach an Alexander type moisture tube (see 6.4) and fill the graduated part of the tube with chloroform. Attach a 40 centimeter Allihn condenser (protected from the atmosphere by a calcium chloride tube) and heat the flask on an electric hot plate for 3 hours in such a way that the distillate falls from the end of the condenser into the graduated part of the moisture tube at the rate of 2 or 3 drops per second. Midway in the 3=hour period, wash down the condenser with four 5=milliliter portions of chloroform. Do not wash again since the washing operation will drive over some water from the moisture tube into the flask. Shut off the heat and read the point of contact of the top meniscus with the wall of the tube and the highest part of the lower meniscus. The difference is water. Calculate as follows:

$$\text{Percent water} = \frac{\text{ml of water x 100}}{\text{gm of sample}}$$

4.4.4.2 *Type II.* Proceed as with type I but weigh a 75-gram sample into a 1-liter balloon flask (this large flask is needed because type II calcium resinate is very bulky).

4.4.5 *Flame test.* Ignite a 1 or 2 gram portion of the material in a porcelain crucible until the ash is white or nearly colorless. Cool the residue, moisten with dilute hydrochloric acid, and evaporate to dryness. Dip the loop of a clean platinum wire into the residue, and then hold the loop in the blue flame of a Bunsen burner. Note if the flame test is characteristic of a calcium as indicated by a yellow-red color with no perceptible green coloration indicating the presence of barium compounds.

4.4.6 *Granulation.*

4.4.6.1 *Type I.* Place a 25-gram sample of type I material on a No. 4 U. S. Standard sieve conforming to Specification RR–S–366; cover and place on a brass pan. Set the nest on a holder of a sieve machine and run for 10 minutes. Weigh the amount remaining in the sieve and pan separately and calculate the percentage. The sieve machine used may be a mechanical shaker geared to produce 300 plus or minus 15 gyrations and 152 plus or minus 10 taps of the strike per minute.

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4.4.6.2 *Type II.* Proceed as with type I but use a No. 60 U. S. Standard sieve.

4.5 **Resubmission and retest.** If the sample subject to test fails to pass the tests, the lot shall be rejected subject to reworking and resubmission. The lot shall be considered acceptable provided that a new sample, selected in accordance with 4.2, passes all the tests required by this specification.

5. PREPARATION FOR DELIVERY.

5.1 Packing.

5.1.1 *Level C.* The calcium resinate shall be packed and delivered in polyethylene bags in airtight containers. The air space at the top of the container should be kept at a minimum. Unless otherwise specified in the contract or order, the capacity of a package shall not exceed 300 pounds.

5.2 **Marking.** In addition to any special marking required by the contract or order, shipments shall be marked in accordance with Standard MIL–STD–129.

5.2.1 *Precautionary Marking.* Each container shall be marked 'Flammable' and 'Store in a cool place away from areas of acute fire hazard and open flame.'

6. NOTES

6.1 **Intended use.** Calcium resinate is intended for use in pyrotechnic compositions.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type required (see 1.2 and 3.2).

(c) Selection of packing and packaging.

6.3 Powminco acid washed Gooch asbestos may be obtained from the Arthur H. Thomas Co, Philadelphia, Pa.

6.4 Alexander-type moisture tubes are available commercially from Scientific Glass Apparatus Co., Vineland, N. J. (Catalogue J–52 and No. J–2078) and Kontes Glass Works, Vineland, N. J. (No. K–75500).

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